Functions, Bijections, and Inverses

MTH 317H — Week 6

Name: _____

Problems 1–2 finish the function problems from the recent Week 5 worksheet. Problems 3–7 develop this week's material. Problems 8–9 are optional review problems.

Recall. Injective means equal outputs force equal inputs. Surjective means every element of the codomain has a preimage. Bijective means both. Justify every claim using the stated domain and codomain.

1. Carryover: disproving injectivity

Prove or disprove: $f\colon \mathbb{R} \to \mathbb{R}$, defined by $f(x) = x^3 - x$, is injective. If you disprove it, exhibit two distinct inputs with the same output.

2. Carryover: proving surjectivity

Prove that $g\colon \mathbb{R}^2 \to \mathbb{R}^2$, defined by

$$g(x, y) = (x + y, x - y),$$

is surjective. Start with an arbitrary target $(u, v) \in \mathbb{R}^2$, find an input (x, y), and check it.

After completing the original problem: Is g also injective? If so, give its inverse and verify both compositions.

3. Bijections and inverses

Let $f\colon X \to Y$. An inverse for f is a function $h\colon Y \to X$ satisfying $h(f(x)) = x$ for all $x \in X$ and $f(h(y)) = y$ for all $y \in Y$. Prove that f is bijective if and only if it has an inverse.

(a) Assume an inverse h exists. Prove that f is injective.

(b) Under the same assumption, prove that f is surjective.

(c) Conversely, assume that f is bijective. Explain why every $y \in Y$ has exactly one preimage. Use this to define h and verify both identities.

Keep track: Which identity is used in (a), and which in (b)?

4. A bijection of the plane

Let $F\colon \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $F(x, y) = (x + y, x + 2y)$. Show that F is bijective by finding an inverse. State the domain and codomain of F^{-1} and check $F^{-1} \circ F$ and $F \circ F^{-1}$.

5. An inverse with excluded values

Consider

$$r\colon \mathbb{R} \setminus \{3\} \longrightarrow \mathbb{R} \setminus \{1\}, \qquad r(x) = \frac{x+1}{x-3}.$$

(a) Check that this formula defines a function with the stated codomain.

(b) Find a candidate inverse and state its domain and codomain.

(c) Check that the candidate has values in its claimed codomain and verify both compositions. Conclude that r is bijective.

6. Intervals and cardinality

Two sets have the same cardinality if there is a bijection between them. Let $a, b \in \mathbb{R}$ with $a < b$.

(a) Prove that $L\colon (0,1) \to (a,b)$, $L(t) = a + (b-a)t$, is bijective by finding and checking its inverse. Check the interval restrictions.

(b) Find a bijection $H\colon (0,\infty) \to (0,1)$ and verify your answer. *Hint:* Try $H(x) = x/(1+x)$.

(c) Use (a) and (b) to construct a bijection from $(0,\infty)$ to (a,b). Give its inverse and state the resulting equality of cardinalities.

7. Composition of bijections

Suppose $f\colon X \to Y$ and $g\colon Y \to Z$ are bijective.

(a) Prove directly that $g \circ f\colon X \to Z$ is injective.

(b) Prove directly that $g \circ f$ is surjective.

(c) Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ by checking both compositions. Explain why the order of the inverses is reversed.

8. Optional review problems: a piecewise integer function

Let $p\colon \mathbb{Z} \to \mathbb{Z}$ be given by

$$p(n) = \begin{cases} n/2, & n \text{ even}, \\ (n+1)/2, & n \text{ odd}. \end{cases}$$

Is p injective? Prove your answer.

9. Optional review problems: an integer-valued function

Consider $q\colon \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $q(m,n) = 3m - 4n$.

(a) Prove that q is a function.

(b) Is q injective? Prove your answer.

(c) Is q surjective? Prove your answer.